



Merewether High School

2010

Higher School Certificate
Trial Examination

Mathematics Extension 2

General Instructions

- Reading time - 5 minutes
- Working time – 3 hours
- Board-approved calculators may be used
- Write using black or blue pen
- A table of standard integrals is provided
- All necessary working should be shown in every question
- Write your student number at the top of every page
- Start each question on a new sheet of paper
- Each question is to be handed in separately

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value

This paper MUST NOT be removed from the examination room

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Question 1**15 Marks****Start a new page**

a.) Find $\int \frac{dx}{x \ln x}$ [2]

b.) Find $\int \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx$ [3]

c.) Evaluate in simplest exact form $\int_0^{\frac{\pi}{3}} \frac{\sec x + \tan x}{\cos x} dx$ [3]

d.) Using partial fractions evaluate $\int_1^2 \frac{3}{3x - x^2} dx$ [3]

e.) Evaluate $\int_{-1}^0 \frac{(x-1)}{x^2 + 2x + 2} dx$ [4]

Question 2**15 Marks****Start a new page**

a.) The complex number z is given by $z = 1 + \frac{1+i}{1-i}$ find:

i. $Re(z)$ [1]

ii. $Im(z)$ [1]

iii. $|z|$ [1]

iv. $\arg(z)$ [1]

b.) On separate Argand diagrams, draw neat labelled sketches for each of the regions:

i. $|z - 2 - i| = 4$ [1]

ii. $Re(z + iz) \geq 2$ [2]

c.) i. Determine the locus of the complex number z given:

$\arg(z - 2) = \frac{\pi}{4} + \arg(z + 2)$ [2]

ii. sketch the locus on the Argand diagram [2]

d.) i. Express $2\sqrt{3} - 2i$ in polar form [1]

ii. Hence or otherwise evaluate $(2\sqrt{3} - 2i)^7$ giving your answer as a complex number in Cartesian form [3]

Question 3

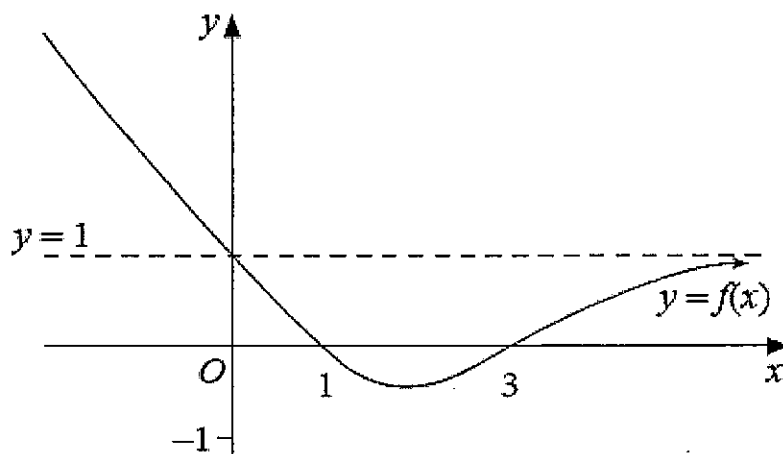
15 Marks

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- a.) The polynomial $P(x) = x^3 - 12x^2 + 36x + c$ has a double zero. Find any possible values of the real number c .

[3]

b.)



The diagram shows the graph of the function $y = f(x)$.

The function has a horizontal asymptote as $y = 1$.

Draw separate half-page sketches of the graphs of the following functions:

i. $y = |f(x)|$ [2]

ii. $y = \frac{1}{f(x)}$ [2]

iii. $y = \ln [f(x)]$ [2]

- c.) $P(x) = x^4 - 2x^3 + 4x^2 - 3x + 1$ and the equation $P(x) = 0$ has roots α, β, γ , and δ .

i. Show that the equation $P(x) = 0$ has no integer roots [1]

ii. Show that $P(x) = 0$ has a real root between 0 and 1 [1]

iii. Show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = -4$ [2]

iv. Hence find the number of real roots of the equation $P(x) = 0$, giving reasons [2]

Question 4**15 Marks****Start a new page**a.) For the curve $y^3 + 2xy + x^2 + 2 = 0$ i. Show that $\frac{dy}{dx} = \frac{-2(y+x)}{3y^2+2x}$ [3]

ii. Find the coordinates of any stationary points on the curve [2]

b.) Points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \varphi, b \sin \varphi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.i. Find the equation of the chord PQ [1]ii. Hence show that if PQ subtends a right angle at the point $A(a, 0)$ then PQ passes through a fixed point $T(t, 0)$ on the x-axis, where $t = \frac{ae^2}{2te^2}$ [4]c.) Find $\int \sin^{-1} x \, dx$ [2]d.) The point A represents the complex number z_3 and the point Z_1 represents the complex number z_1 . The point Z_1 is rotated about A through a right angle in the positive direction to take up the position Z_2 , representing the complex number z_2 . Show that $z_2 = (1 - i)z_3 + iz_1$ [3]**Question 5****15 Marks****Start a new page**

a.) Use Mathematical Induction to show that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

for all positive integers $n \geq 2$ [4]

b.) The area below the curve $y = bx - ax^2$ (where $a > 0$ and $b > 0$) and above the x-axis is rotated about the y axis through a complete revolution.

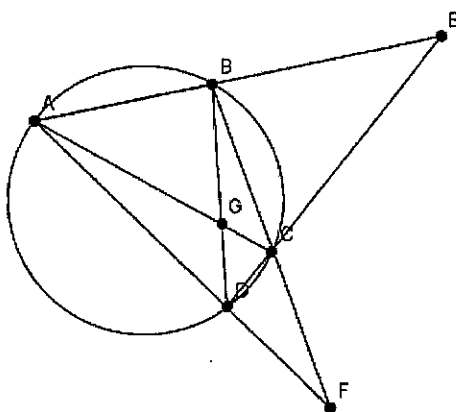
Show, using a slice technique or otherwise, that the volume of the solid so formed is:

$$\frac{\pi b^4}{6a^3} \text{ cubic units} \quad [3]$$

- c.) Patrick's accuracy with darts is such that if he scores a bullseye, the probability of doing the same on the next throw is $\frac{2}{3}$, however if he misses, the probability that he again misses the bullseye on the next throw is $\frac{3}{4}$.
The probability of hitting the bullseye on the first throw is $\frac{1}{3}$.
- What is the probability that he throws a bullseye on the second throw? [2]
 - What is the probability that he misses a bullseye on the third throw? [2]
- d.) On a certain day, the depth of water in a harbour at high tide at 5 am is 9 m. At the following low tide at 11:20 am the depth is 3 m. Assuming that the tidal motion is simple harmonic, find the latest time before noon that a ship can enter the harbour if a minimum depth of 7.5 m of water is required. [4]

Question 6**15 Marks****Start a new page**

- a.) For the hyperbola $\frac{x^2}{16} - \frac{y^2}{128} = 1$ find:
- The eccentricity [1]
 - The coordinates of the foci [1]
 - The equations of the directrices [1]
- b.) ABCD is a cyclic quadrilateral whose opposite sides meet at E and F and whose diagonals meet at G.

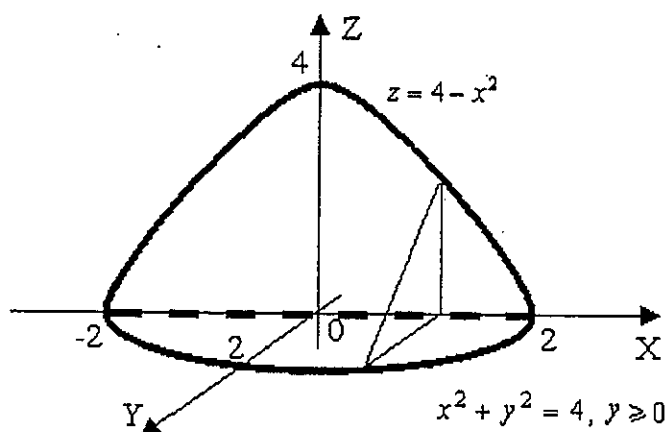


- If BD bisects the angles at B and D, prove that $\angle BAD$ is a right angle. [3]
- What is the relation between the angles EAF and ECF? [3]

- c.) By taking strips parallel to the axis of rotation, use the method of cylindrical shells to find the volume of the solid obtained by rotating the region $\{(x, y): 0 \leq y \leq 2x - x^2\}$ about the y -axis. [3]
- d.) By using two applications of integration by parts, evaluate $\int_0^{\frac{\pi}{2}} e^x \cos x \, dx$ [3]

Question 7**15 Marks****Start a new page**

- a.) The solid shown has a semicircular base of radius 2 units. Vertical cross-sections perpendicular to the diameter are right-angled triangles whose height is bounded by the parabola $z = 4 - x^2$.



- i. By slicing at right angles to the x -axis, show that the volume of the solid is given by $V = \int_0^2 (4 - x^2)^{3/2} \, dx$, [3]
- ii. and hence calculate this volume. [3]
- b.) For positive real numbers $a, b, c, a_1, a_2, \dots, a_n$:
- i. Show that $a + \frac{1}{a} \geq 2$ [1]
- ii. Hence show that $(a + b) \left(\frac{1}{a} + \frac{1}{b} \right) \geq 4$
and $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$ [2]
- iii. Show that $(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \geq n^2$ [2]

- c.) $P\left(cp, \frac{c}{p}\right), Q\left(cq, \frac{c}{q}\right), R\left(cr, \frac{c}{r}\right)$ are three points on the rectangular hyperbola $xy = c^2$ such that the parameters p, q, r are in geometric progression.
- Explain why P and R must lie on the same branch of the hyperbola.
Under what condition will Q lie on the opposite branch to P and R ? [1]
 - Show that the chord PR is parallel to the tangent to the hyperbola at Q . [3]

Question 8**15 Marks****Start a new page**

- a.) $I_n = \int_0^1 \ln(1+x) dx, n = 0, 1, 2, \dots$
- Show that $\int \ln(1+x) dx = (1+x)\ln(1+x) - x + c$ [1]
 - Show that $(n+1)I_n = 2\ln 2 - \frac{1}{n+1} - nI_{n-1}, n = 1, 2, \dots$ [2]
 - Evaluate $3I_2$ and $4I_3$ [2]
 - Show that $(n+1)I_n = \begin{cases} \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}, & n \text{ odd} \\ 2\ln 2 - \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}\right), & n \text{ even} \end{cases}$ [2]
- b.) i. If $t = \tan x$ prove that $\tan 4x = \frac{4t(1-t^2)}{t^4-6t^2+1}$ [2]
- If $\tan x \tan 4x = 1$ deduce that $5t^4 - 10t^2 + 1 = 0$ [1]
 - Prove that $x = 18^\circ$ and $x = 54^\circ$ satisfy the equation $\tan x \tan 4x = 1$ [2]
 - Deduce that $\tan 54^\circ = \sqrt{\frac{5+2\sqrt{5}}{5}}$ [3]

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Extension 2 Trial 2010

$$1) a) \int \frac{dx}{x \ln x} = I \quad \text{Let } u = \ln x \\ du = \frac{1}{x} \cdot dx \quad (1)$$

$$I = \int \frac{du}{u} \\ = \ln u + C \\ = \ln(\ln x) + C \quad (1)$$

$$b) \int \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx = I \\ \text{Let } 2y = \sin x \\ 2dy = \cos x dx$$

$$I = \int \frac{2dy}{\sqrt{4 - 4y^2}} \quad (1)$$

$$= \int \frac{dy}{\sqrt{1 - y^2}}$$

$$= \sin^{-1} y + C \quad (1)$$

$$= \sin^{-1} \left(\frac{\sin x}{2} \right) + C \quad (1)$$

$$c) \int_0^{\frac{\pi}{3}} \frac{\sec x + \tan x}{\cos x} dx \\ = \int_0^{\frac{\pi}{3}} (\sec^2 x + \sec x \tan x) dx \quad (1)$$

$$= [\tan x + \sec x]_0^{\frac{\pi}{3}} \quad (1)$$

$$= (\sqrt{3} + 2) - (0 + 1)$$

$$= \sqrt{3} + 1 \quad (1)$$

$$d) I = \int_1^2 \frac{3}{3x - x^2} dx$$

$$\frac{3}{3x - x^2} = \frac{A}{3 - x} + \frac{B}{x}, \quad x \neq 0, 3$$

$$3 = Ax + 3B - Bx$$

$$3 = (A - B)x + 3B$$

$$\therefore B = 1$$

$$A - B = 0$$

$$\therefore A = 1 \quad (1)$$

$$I = \int_1^2 \frac{1}{3 - x} + \frac{1}{x} dx$$

$$= \left[-\ln |3 - x| + \ln |x| \right]_1^2 \quad (1)$$

$$= (-\ln(1) + \ln 2) - (-\ln(2) + \ln 1)$$

$$= 2 \ln(2) \quad (1)$$

$$e) I = \int_{-1}^0 \frac{(x-1)}{x^2 + 2x + 2} dx$$

$$= \frac{1}{2} \int_{-1}^0 \frac{(2x-2)}{x^2 + 2x + 2} dx$$

$$= \frac{1}{2} \int_{-1}^0 \frac{2x+2}{x^2 + 2x + 2} dx - \frac{1}{2} \int_{-1}^0 \frac{4}{x^2 + 2x + 2} dx \quad (1)$$

$$= \frac{1}{2} \left[\ln |x^2 + 2x + 2| \right]_{-1}^0 - \int_{-1}^0 \frac{2}{(x-1)^2 + 1} dx \quad (1)$$

$$= \frac{1}{2} (\ln 2 - \ln 1) - 2 \left[\tan^{-1}(x+1) \right]_{-1}^0 \quad (1)$$

$$= \frac{1}{2} \ln 2 - 2 (\tan^{-1} 1 - \tan^{-1} 0)$$

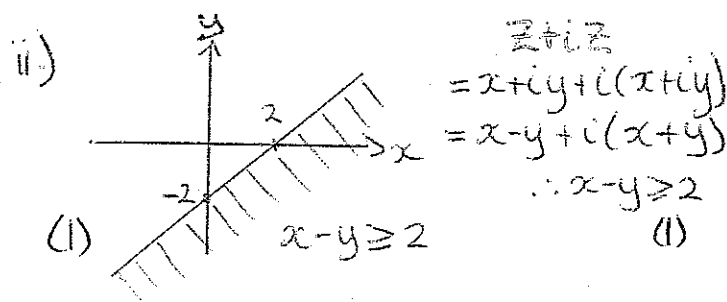
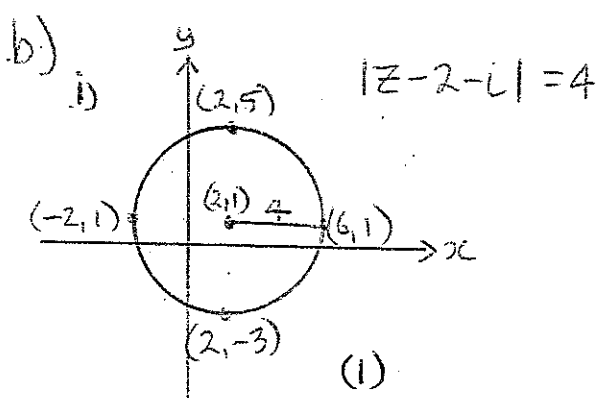
$$= \frac{1}{2} \ln 2 - 2 \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{1}{2} \ln 2 - \frac{\pi}{2} \quad (1)$$

Question 2.

$$\begin{aligned} \text{a.) } Z &= 1 + \frac{1+i}{1-i} \\ &= 1 + \frac{(1+i)^2}{(1-i)(1+i)} \\ &= \frac{1+i^2-1}{2} \\ &= 1+i \end{aligned}$$

- i) $\operatorname{Re}(Z) = 1$ (1)
 ii) $\operatorname{Im}(Z) = 1$ (1)
 iii) $|Z| = \sqrt{2}$ (1)
 iv) $\arg Z = \tan^{-1} \frac{1}{1} = \frac{\pi}{4}$ (1)



$$\operatorname{Re}(Z+iZ) \geq 2$$

$$\begin{aligned} \text{c.) i) } \arg(Z-2) &= \frac{\pi}{4} + \arg(Z+2) \\ \text{Let } \arg(Z-2) &= \theta \\ \text{and } \arg(Z+2) &= \phi \\ \therefore \theta &= \frac{\pi}{4} + \phi \end{aligned}$$

$$\tan \theta = \tan\left(\frac{\pi}{4} + \phi\right)$$

$$= \frac{\tan \frac{\pi}{4} + \tan \phi}{1 - \tan \frac{\pi}{4} \tan \phi}$$

$$= \frac{1 + \tan \phi}{1 - \tan \phi} \quad (1)$$

$$Z-2 = (x-2) + iy$$

$$\therefore \tan \theta = \frac{y}{x-2}$$

and

$$Z+2 = (x+2) + iy$$

$$\therefore \tan \phi = \frac{y}{x+2}$$

$$\therefore \frac{y}{x-2} = \frac{1 + \frac{y}{x+2}}{1 - \frac{y}{x+2}}$$

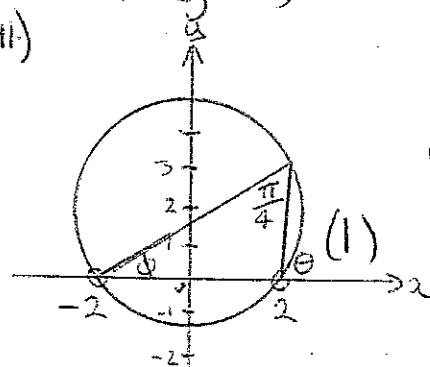
$$\frac{y}{x-2} = \frac{x+2+y}{x+2-y}$$

$$xy + zy - y^2 = (x^2 - 4) + xy - 2y$$

$$x^2 + y^2 - 4y - 4 = 0$$

$$x^2 + (y-2)^2 = 8 \quad (1)$$

ii)



Consider the sketch centre $(0,2)$, radius $2\sqrt{2}$

From the sketch, a restriction is put on the locus and its equation becomes $x^2 + (y-2)^2 = 8, y \geq 0$ (1)

$$2d) 2\sqrt{3} - 2i = r \operatorname{cis} \theta$$

$$\begin{aligned} \text{Then } r &= \sqrt{(2\sqrt{3})^2 + (-2)^2} \\ &= \sqrt{16} \\ &= 4 \end{aligned}$$

$$\tan \theta = \frac{-2}{2\sqrt{3}}$$

$$= -\frac{1}{\sqrt{3}}$$

$$\therefore \theta = -\frac{\pi}{6}$$

$$\therefore 2\sqrt{3} - 2i = 4 \operatorname{cis}\left(-\frac{\pi}{6}\right) \quad (1)$$

$$\begin{aligned} \text{ii) } (2\sqrt{3} - 2i)^7 &= \left[4 \operatorname{cis}\left(-\frac{\pi}{6}\right)\right]^7 \\ &= 4^7 \operatorname{cis}\left(-\frac{7\pi}{6}\right) \end{aligned}$$

$$= 16384 \operatorname{cis}\left(\frac{5\pi}{6}\right) \quad (1)$$

$$= 16384 \left[\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right) \right]$$

$$= 16384 \left[-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right] \quad (1)$$

$$= -8192\sqrt{3} + 8192i \quad (1)$$

Question 5

$$a) P(x) = x^3 - 12x^2 + 36x + C$$

$$P'(x) = 3x^2 - 24x + 36$$

$$= 3(x-6)(x-2)$$

$$\therefore P'(x) = 0 \text{ for } x = 6, 2 \quad (1)$$

$$\text{So } P'(2) = P(2) = 0$$

$$\text{Hence } P(2) = 2^3 - 12(2)^2 + 36(2) + C = 0$$

$$\therefore 8 - 48 + 72 = -C$$

$$\therefore C = -32 \quad (1)$$

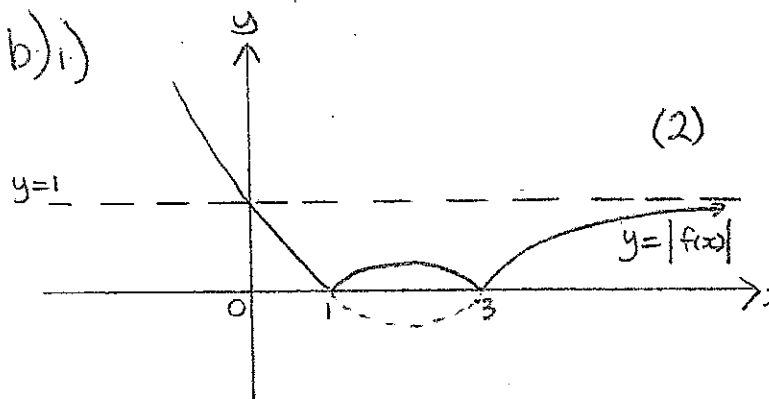
$$\text{Also } P'(6) = P(6) = 0$$

$$\text{Hence } P(6) = 6^3 - 12(6^2) + 36(6) + C = 0$$

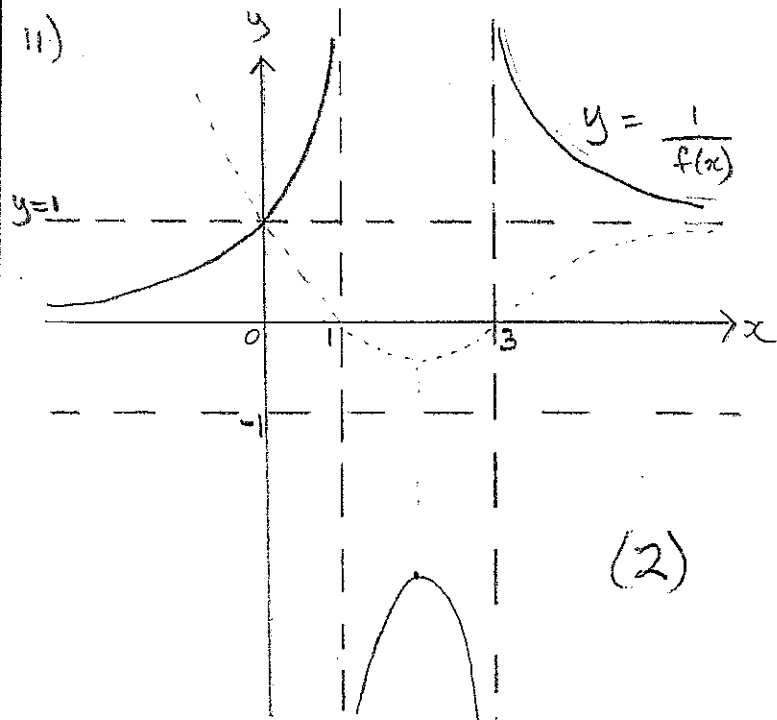
$$\therefore C = 0 \quad (1)$$

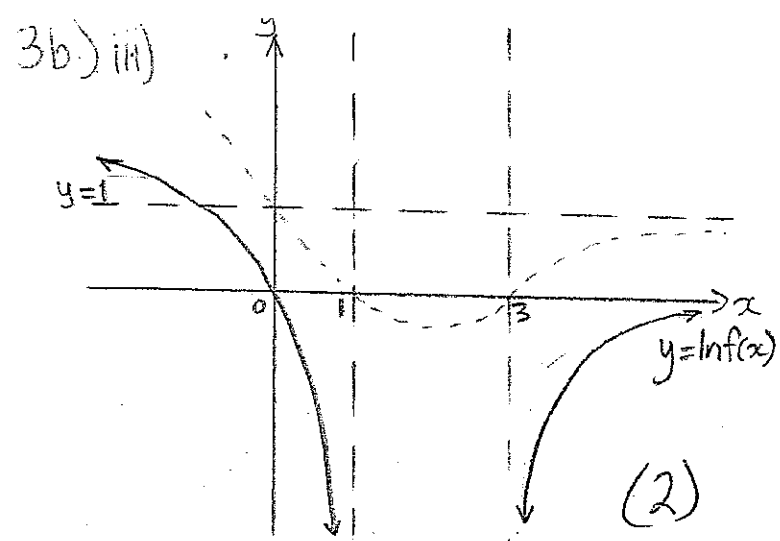
$\therefore P(x)$ has a double zero if and only if $C = -32$ or $C = 0$

b) i)



ii)





c.) $P(x) = x^4 - 2x^3 + 4x^2 - 3x + 1$
 α, β, γ and δ are the roots of $P(x) = 0$

i) Only possible integer roots are ± 1

But $P(1) = 1 - 2 + 4 - 3 + 1$
 $= -1$
 $\neq 0$

$P(-1) = 1 + 2 + 4 + 3 + 1$
 $= 11$
 $\neq 0$ (1)

Hence there are no integer roots

ii) $P(x)$ is a continuous, real function and $P(0) = 1 > 0$ also $P(1) = -1 < 0 \therefore$ indeterminate.

~~Hence, considering the graph of $y = P(x)$, a real root exists between 0 and 1~~

ii) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$
 $(\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$
 $= 2^2 - 2 \times 4$ (1)
 $= -4$

iv) Since $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = -4$, at least one of these squares must be negative.
Hence $P(x) = 0$ has a non-real root; then its complex conjugate is a second non-real root, since the coefficients of $P(x)$ are real. (1)

We know there is a real root between 0 and 1. Since the non-real roots come in complex conjugate pairs, the remaining fourth root cannot be non-real. (1)

Hence the equation $P(x) = 0$ has two real roots and two non-real roots.

Question 4

a) $y^3 + 2xy + x^2 + 2 = 0$

i) $3y^2 \frac{dy}{dx} + 2(1 \cdot y + x \frac{dy}{dx}) + 2x = 0$ (1)

$\frac{dy}{dx} (3y^2 + 2x) = -2(y + x)$ (1)

$\frac{dy}{dx} = \frac{-2(y+x)}{3y^2 + 2x}$ (1)

ii) $\frac{dy}{dx} = 0$ for $y = -x$ and

$y^3 + 2xy + x^2 + 2 = 0$
(sub in $y = -x$) (1)

$-x^3 - 2x^2 + x^2 + 2 = 0$

$-x^3 - x^2 + 2 = 0$

$(x-1)(x^2 + 2x + 2) = 0$

Hence $(1, -1)$ is the only stationary point since quadratic factor has $\Delta < 0$ (1)

4b)i)

$P(a \cos \theta, b \sin \theta)$, $Q(a \cos \phi, b \sin \phi)$ ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

gradient of PQ is

$$\begin{aligned} \frac{y_2 - y_1}{x_2 - x_1} &= \frac{b(\sin \theta - \sin \phi)}{a(\cos \theta - \cos \phi)} = \frac{b}{a} \cdot \frac{2 \sin\left(\frac{\theta - \phi}{2}\right) \cos\left(\frac{\theta + \phi}{2}\right)}{-2 \sin\left(\frac{\theta - \phi}{2}\right) \sin\left(\frac{\theta + \phi}{2}\right)} \\ &= -\frac{b}{a} \frac{\cos\left(\frac{\theta + \phi}{2}\right)}{\sin\left(\frac{\theta + \phi}{2}\right)} \end{aligned}$$

$\therefore$ the equation of chord PQ is

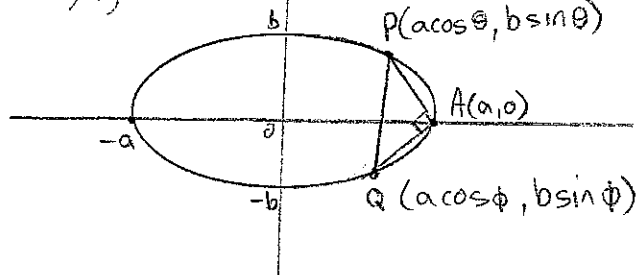
$$y - b \sin \theta = -\frac{b}{a} \frac{\cos\left(\frac{\theta + \phi}{2}\right)}{\sin\left(\frac{\theta + \phi}{2}\right)} (x - a \cos \theta)$$

$$\begin{aligned} \frac{x}{a} \cos\left(\frac{\theta + \phi}{2}\right) + \frac{y}{b} \sin\left(\frac{\theta + \phi}{2}\right) &= \cos \theta \cos\left(\frac{\theta + \phi}{2}\right) \\ &\quad + \sin \theta \sin\left(\frac{\theta + \phi}{2}\right) \end{aligned}$$

$$\frac{x}{a} \cos\left(\frac{\theta + \phi}{2}\right) + \frac{y}{b} \sin\left(\frac{\theta + \phi}{2}\right) = \cos\left(\frac{\theta - \phi}{2}\right)$$

where P, Q have parameters θ, ϕ .

4. h) i)



The chord PQ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has the eqⁿ

$$\frac{x}{a} \cos\left(\frac{\theta+\phi}{2}\right) + \frac{y}{b} \sin\left(\frac{\theta+\phi}{2}\right) = \cos\left(\frac{\theta-\phi}{2}\right)$$

where P, Q have parameters θ, ϕ .
(1)

ii) The chord PQ cuts the x-axis at point T(t, 0).

$$t = a \cos\left(\frac{\theta-\phi}{2}\right) \sec\left(\frac{\theta+\phi}{2}\right)$$

$$= a \left(1 + \tan\frac{\theta}{2} \tan\frac{\phi}{2}\right) \left(1 - \tan\frac{\theta}{2} \tan\frac{\phi}{2}\right)^{-1}$$

The gradient AP is $\frac{b \sin \theta}{a(\cos \theta - 1)}$

$$= -\frac{b}{a} \cot \frac{\theta}{2}$$

and gradient AQ is $\frac{b \sin \phi}{a(\cos \phi - 1)}$

$$= -\frac{b}{a} \cot \frac{\phi}{2}$$

If the chord PQ subtends a right angle at the point A, then $AP \times \text{gradient } AQ = -1$.

Therefore

$$\frac{b^2}{a^2} \cot \frac{\theta}{2} \cot \frac{\phi}{2} = -1$$

$$\tan \frac{\theta}{2} \tan \frac{\phi}{2} = -\frac{b^2}{a^2} \quad (1)$$

Hence $t = a \left(1 - \frac{b^2}{a^2}\right) \left(1 + \frac{b^2}{a^2}\right)^{-1}$

$$= a \left(\frac{a^2 - b^2}{a^2 + b^2}\right) \quad (1)$$

But for the ellipse

$$b^2 = a^2(1 - e^2)$$

thus $t = \frac{ae^2}{2 + e^2} \quad (1)$

So PQ passes through a fixed point $T\left(\frac{ae^2}{2 + e^2}, 0\right)$ on the x-axis.

[4]

$$4) c) \int \sin^{-1} x \, dx \quad \begin{matrix} u=x & v=\sin^{-1} x \\ u'=1 & v'=\frac{1}{\sqrt{1-x^2}} \end{matrix}$$

$$= x \sin^{-1} x - \int x \times \frac{1}{\sqrt{1-x^2}} \, dx$$

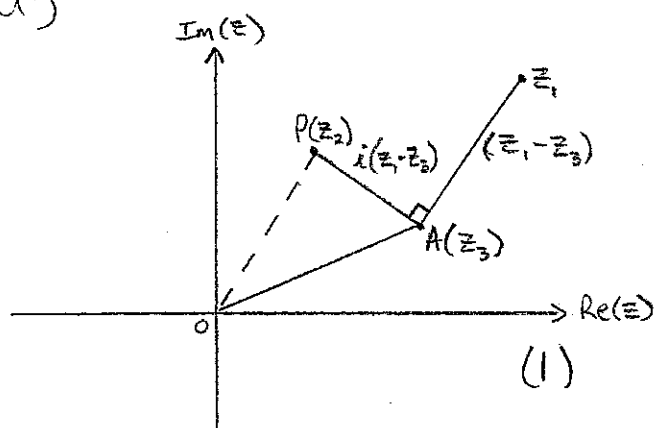
$$= x \sin^{-1} x - \int x (1-x^2)^{-\frac{1}{2}} \, dx \quad (1)$$

$$= x \sin^{-1} x + \frac{1}{2} \int -2x (1-x^2)^{-\frac{1}{2}} \, dx$$

$$= x \sin^{-1} x + \frac{1}{2} \frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= x \sin^{-1} x + \sqrt{1-x^2} + C \quad (1)$$

d)



$$\vec{AZ_1} = z_1 - z_3$$

$$\vec{AP} = i(z_1 - z_3)$$

$$\vec{OP} = \vec{OA} + \vec{AP} \quad (1)$$

$$\therefore OP = z_3 + i(z_1 - z_3)$$

$$= z_3 + i z_1 - i z_3 \quad (1)$$

$$= (1-i) z_3 + i z_1$$

Question 5

a) Let $s(n)$, $n=2,3,\dots$ be the sequence defined by

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

Consider $s(2)$:

$$\frac{1}{1^2} + \frac{1}{2^2} = \frac{5}{4} < \frac{6}{4} = 2 - \frac{1}{2}$$

Hence true for $s(2)$ (1)

Assume true for $s(k)$:

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} < 2 - \frac{1}{k} \quad (1)$$

Show true for $s(k+1)$ if true for $s(k)$:

$$\text{ie } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \frac{1}{(k+1)}$$

$$\text{RHS} = 2 - \frac{1}{k} + \frac{1}{(k+1)}$$

$$= 2 - \frac{1}{k+1} \left(\frac{k+1}{k} - \frac{1}{k+1} \right)$$

$$= 2 - \frac{1}{k+1} \left(\frac{(k+1)^2 - k}{k(k+1)} \right)$$

$$= 2 - \frac{1}{k+1} \left(\frac{k^2 + 2k + 1 - k}{k(k+1)} \right)$$

$$= 2 - \frac{1}{k+1} \left(\frac{k^2 + k + 1}{k(k+1)} \right)$$

$$= 2 - \frac{1}{k+1} \left(\frac{k(k+1)}{k(k+1)} + \frac{1}{k(k+1)} \right)$$

$$= 2 - \frac{1}{k+1} \left(1 + \frac{1}{k(k+1)} \right)$$

$$< 2 - \frac{1}{k+1} \quad \text{since } k(k+1) > 0 \quad (1)$$

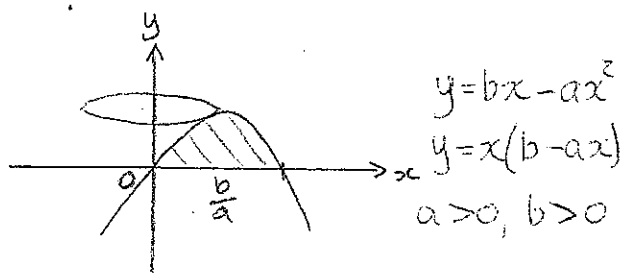
$$\therefore \text{LHS} < 2 - \frac{1}{k+1}$$

$\therefore s(k+1)$ is true if $s(k)$ is true.

But $s(2)$ is true $\therefore s(3)$ is true...

Hence $s(n)$ is true for $n=2,3,4,\dots$ (1)

5) b.)



$$\begin{aligned}
 V &= \int_0^{\frac{b}{a}} 2\pi xy \, dx \\
 &= \int_0^{\frac{b}{a}} 2\pi \cdot x \cdot (bx - ax^2) \, dx \quad (1) \\
 &= 2\pi \int_0^{\frac{b}{a}} (bx^2 - ax^3) \, dx \\
 &= 2\pi \left[\frac{bx^3}{3} - \frac{ax^4}{4} \right]_0^{\frac{b}{a}} \quad (1) \\
 &= 2\pi \left[\frac{b}{3} \times \frac{b^3}{a^3} - \frac{a}{4} \times \frac{b^4}{a^4} - 0 \right] \\
 &= 2\pi \left[\frac{b^4}{3a^3} - \frac{b^4}{4a^3} \right] \\
 &= 2\pi \frac{b^4}{12a^3} \\
 &= \frac{\pi b^4}{6a^3} \text{ cubic units} \quad (1)
 \end{aligned}$$

c)

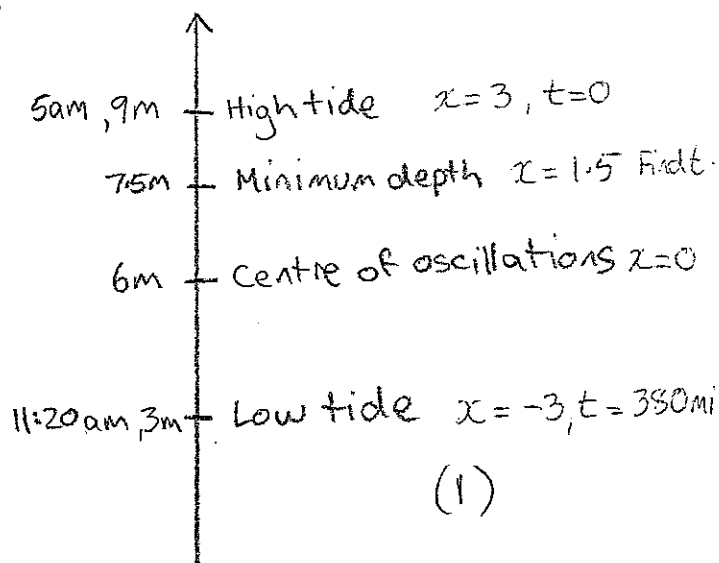
i) $P(\text{B on second})$

$$\begin{aligned}
 &= \frac{1}{3} \times \frac{2}{3} + \frac{2}{3} \times \frac{1}{4} \quad (1) \\
 &= \frac{2}{9} + \frac{1}{6} \\
 &= \frac{7}{18} \quad (1)
 \end{aligned}$$

11) $P(\text{Misses on 3rd}) =$

$$\begin{aligned}
 &P(B, B, NB) + P(B, NB, NB) + P(NB, NB, NB) + P(NB, B, NB) \\
 &= \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} \times \frac{3}{4} + \frac{2}{3} \times \frac{3}{4} \times \frac{3}{4} + \frac{2}{3} \times \frac{1}{4} \times \frac{1}{3} \quad (1) \\
 &= \frac{2}{27} + \frac{1}{12} + \frac{3}{8} + \frac{1}{18} \\
 &= \frac{127}{216} \text{ or } 0.58796 \quad (1)
 \end{aligned}$$

d)



Period $T = 2(11:20 - 5) = 760$ minutes.

Amplitude $= \frac{1}{2}(9 - 3) = 3\text{m}$

Motion is SHM: $\ddot{x} = -n^2x$

$$n = \frac{2\pi}{T} = \frac{\pi}{380} \quad (1)$$

$\therefore x = 3 \cos(nt + \alpha)$, $0 \leq \alpha < 2\pi$

Initial conditions: $t = 0$, $x = 3$, $\cos \alpha = 1$
 $\alpha = 0 \therefore x = 3 \cos nt$ (1)

A minimum depth is 7.5m if $x = 1.5$

$$1.5 = 3 \cos\left(\frac{\pi}{380} t\right)$$

$$\frac{1}{2} = \cos\left(\frac{\pi}{380} t\right)$$

$$\frac{\pi t}{380} = \frac{\pi}{3}$$

$$t = \frac{380}{3} = 2.06$$

Hence the latest time before noon when a minimum depth of 7.5m of water is $5 + 2.06 = 7.06\text{am}$. (1)

QUESTION 6

a) i) $b^2 = a^2(e^2 - 1)$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$= 1 + \frac{128}{16}$$

$$= 9$$

$$\therefore e = 3 \quad (1)$$

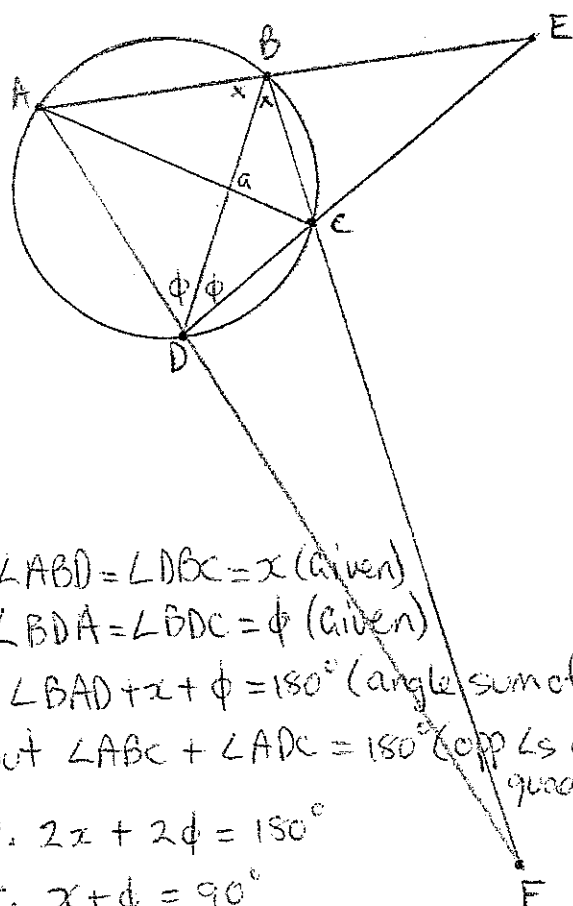
ii) Foci have coordinates $(\pm ae, 0)$

$$\therefore (9, 0) \text{ and } (-9, 0) \quad (1)$$

iii) Directrices have equations $x = \pm \frac{a}{e}$

$$\therefore x = \frac{4}{3} \text{ and } x = -\frac{4}{3} \quad (1)$$

b)



$$\angle ABD = \angle DBC = x \text{ (given)}$$

$$\angle BDA = \angle BDC = \phi \text{ (given)}$$

$$\therefore \angle BAD + x + \phi = 180^\circ \text{ (angle sum of } \Delta = 180^\circ) \quad (1)$$

$$\text{But } \angle ABC + \angle ADC = 180^\circ \text{ (opp } \angle \text{s cyclic quadrangle)} \quad (1)$$

$$\therefore 2x + 2\phi = 180^\circ$$

$$\therefore x + \phi = 90^\circ$$

$$\therefore \angle BAD + 90^\circ = 180^\circ \quad (1)$$

$$\therefore \angle BAD = 90^\circ$$

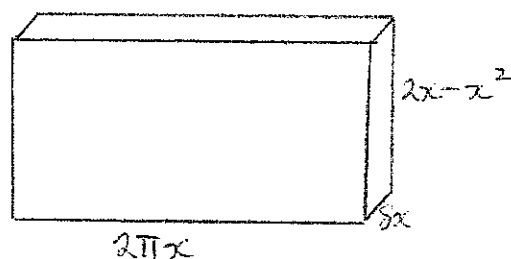
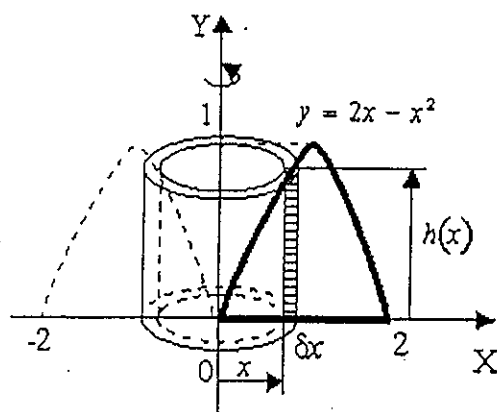
$$\therefore \angle EAF = \angle BAD = 90^\circ$$

$$\angle BCD = 90^\circ \text{ since } \angle BAD + \angle BCD = 180^\circ \text{ (cyclic quad)} \quad (1)$$

$$\therefore \angle ECF = \angle BCD \text{ (vert opp } \angle \text{s equal)} \quad (1)$$

$$\therefore \angle EAF = \angle ECF = 90^\circ \quad (1)$$

c)



Take a typical cylindrical shell of height $2x - x^2$

inner radius x

outer radius $x + \delta x$

The shell has volume

$$\delta V = \pi [(x + \delta x)^2 - x^2] (2x - x^2) \quad (1)$$

$$= 2\pi x (2x - x^2) \delta x \text{ (ignoring } (\delta x)^2)$$

$$\therefore \text{Total } V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 2\pi x (2x - x^2) \delta x \quad (1)$$

$$= 2\pi \int_0^2 x (2x - x^2) dx$$

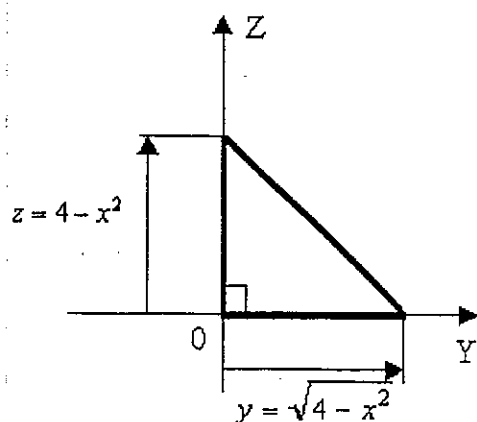
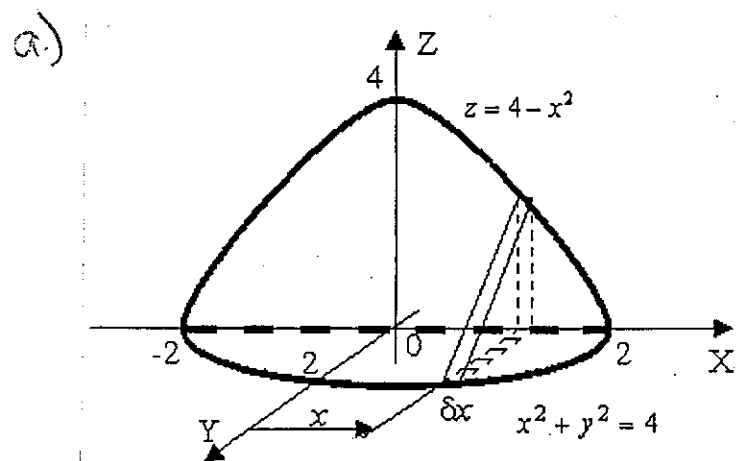
$$= 2\pi \left[2 \frac{x^3}{3} - \frac{x^4}{4} \right]_0^2$$

$$= 2\pi \left[\left(\frac{16}{3} - 4 \right) - 0 \right]$$

$$= \frac{8\pi}{3} \text{ cubic units} \quad (1)$$

$$\begin{aligned}
 \text{b) d) } I &= \int_0^{\frac{\pi}{2}} e^x \cos x \, dx \\
 &= \left[e^x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} e^x \sin x \, dx \quad (1) \\
 &= \left(e^{\frac{\pi}{2}} \cos \frac{\pi}{2} - e^0 \cos 0 \right) + \int_0^{\frac{\pi}{2}} e^x \sin x \, dx \\
 &= -1 + \left[e^x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x \cos x \, dx \quad (1) \\
 &= -1 + e^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x \cos x \, dx \\
 \therefore 2 \int_0^{\frac{\pi}{2}} e^x \cos x \, dx &= e^{\frac{\pi}{2}} - 1 \\
 \text{so } \int_0^{\frac{\pi}{2}} e^x \cos x \, dx &= \frac{1}{2} (e^{\frac{\pi}{2}} - 1) \quad (1)
 \end{aligned}$$

Question 7



i) The slice is a right-angled triangle with area of cross-section A , thickness δx .

$$\begin{aligned}
 A(x) &= \frac{yz}{2} \\
 &= \frac{(4-x^2)^{\frac{3}{2}}}{2} \quad (1)
 \end{aligned}$$

The slice has volume

$$\begin{aligned}
 \delta V &= A(x) \delta x \\
 &= \frac{(4-x^2)^{\frac{3}{2}}}{2} \delta x \quad (1)
 \end{aligned}$$

The volume of the solid is

$$\begin{aligned}
 V &= \lim_{\delta x \rightarrow 0} \sum_{x=-2}^2 \frac{(4-x^2)^{\frac{3}{2}}}{2} \delta x \\
 &= \frac{1}{2} \int_{-2}^2 (4-x^2)^{\frac{3}{2}} \, dx \quad (1)
 \end{aligned}$$

ii) Let $x = 2 \sin \phi$ $\begin{matrix} x=-2, \phi=0 \\ x=2, \phi=\frac{\pi}{2} \end{matrix}$

$$\begin{aligned}
 dx &= 2 \cos \phi \, d\phi \\
 V &= 16 \int_0^{\frac{\pi}{2}} (1 - \sin^2 \phi)^{\frac{3}{2}} \cos \phi \, d\phi \quad (1) \\
 &= 16 \int_0^{\frac{\pi}{2}} \cos^4 \phi \, d\phi \\
 &= 16 \int_0^{\frac{\pi}{2}} \left[\frac{1}{2} (1 + \cos 2\phi) \right]^2 d\phi \\
 &= 4 \int_0^{\frac{\pi}{2}} \left[1 + 2 \cos 2\phi + \frac{1}{2} (1 + \cos 4\phi) \right] d\phi \quad (1) \\
 &= 4 \left[\frac{3}{2} \phi + \sin 2\phi + \frac{\sin 4\phi}{8} \right]_0^{\frac{\pi}{2}} \\
 &= 3\pi \quad (1)
 \end{aligned}$$

$\therefore$ the volume of the solid is 3π cubic units.

i) b. i)

$$\left(a + \frac{1}{a}\right)^2 = \left(a - \frac{1}{a}\right)^2 + 4 \geq 4$$

Since $\left(a - \frac{1}{a}\right)$ real

$$\text{then } \left(a - \frac{1}{a}\right)^2 \geq 0$$

$$\therefore \left(a + \frac{1}{a}\right) \geq 2 \text{ for } a > 0 \quad (1)$$

$$\text{ii.) } (a+b)\left(\frac{1}{a} + \frac{1}{b}\right)$$

$$= 1 + 1 + \frac{a}{b} + \frac{b}{a}$$

$$= 2 + \frac{a}{b} + \frac{b}{a} \geq 4,$$

using part i) with $a \rightarrow \frac{a}{b} \quad (1)$

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

$$= 1 + 1 + 1 + \left(\frac{a}{b} + \frac{b}{a}\right) + \left(\frac{b}{c} + \frac{c}{b}\right) + \left(\frac{c}{a} + \frac{a}{c}\right)$$

$$\geq 3 + 3 \times 2$$

$$= 9 \quad (1)$$

$$\text{iii.) } (a_1 + a_2 + \dots + a_n)\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right)$$

$$= \sum_{i=1}^n \frac{a_i}{a_i} + \sum_{\substack{i=1 \\ j>i}}^n \left(\frac{a_i}{a_j} + \frac{a_j}{a_i}\right) \quad (1)$$

$$= n \times 1 + \sum_{\substack{i=1 \\ j>i}}^n \left(\frac{a_i}{a_j} + \frac{a_j}{a_i}\right)$$

There are nC_2 ways of selecting two different integers from $1, 2, 3, \dots, n$ hence there are nC_2 terms of the form $\frac{a_i}{a_j} + \frac{a_j}{a_i}$ where $i < j$.

$$\therefore (a_1 + a_2 + \dots + a_n)\left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right)$$

$$\geq n + {}^nC_2 \times 2$$

$$= n + n(n-1)$$

$$= n^2 \quad (1)$$

$$\text{c) i) } \frac{q}{p} = \frac{r}{q}$$

$$pr = q^2 > 0$$

Hence p and r must have the same sign, and P and R must lie on the same branch. Q will lie on the opposite branch if the common ratio of the GP is negative. (1)

ii) Gradient of PR is

$$\frac{c\left(\frac{1}{r} - \frac{1}{p}\right)}{c(r-p)} = \frac{p-r}{pr(r-p)} = \frac{-1}{pr} \quad (1)$$

$$y = \frac{c}{t}$$

$$x = ct$$

$$\frac{dy}{dt} = \frac{-c}{t^2}$$

$$\frac{dx}{dt} = c$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \frac{-1}{t^2} \quad (1)$$

$\therefore$ tangent at Q has

$$\text{gradient } \frac{-1}{q^2} = \frac{-1}{pr} \quad (1)$$

Hence PR is parallel to tangent at Q .

Question >

$$a) i) \frac{d}{dx} [(1+x) \ln(1+x) - x]$$

$$= 1 \cdot \ln(1+x) + (1+x) \cdot \frac{1}{1+x} - 1$$

$$= \ln(1+x) \quad (1)$$

$$\therefore \int \ln(1+x) dx = (1+x) \ln(1+x) - x + C$$

$$ii) I_n = \int_0^1 x^n \ln(1+x) dx, \quad n=0, 1, 2, \dots$$

$$= [x^n ((1+x) \ln(1+x) - x)]_0^1$$

$$- n \int_0^1 x^{n-1} [(1+x) \ln(1+x) - x] dx,$$

$$n=1, 2, \dots \quad (1)$$

$$= 2 \ln 2 - 1 - n \int_0^1 [x^{n-1} \ln(1+x) + x^n \ln(1+x) - x^n] dx$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + \frac{n}{n+1} [x^{n+1}]_0^1$$

$$= 2 \ln 2 - 1 - n I_{n-1} - n I_n + \frac{n}{n+1}$$

$$= 2 \ln 2 - n I_{n-1} - n I_n + \frac{1}{n+1}$$

$$\therefore (n+1) I_n = 2 \ln 2 - \frac{1}{n+1} - n I_{n-1},$$

$$n=1, 2, \dots$$

$$ii) I_0 = \int_0^1 \ln(1+x) dx$$

$$= [(1+x) \ln(1+x) - x]_0^1$$

$$= 2 \ln 2 - 1$$

$$3 I_2 = 2 \ln 2 - \frac{1}{3} - 2 I_1$$

$$= 2 \ln 2 - \frac{1}{3} - (2 \ln 2 - \frac{1}{2} - I_0)$$

$$3 I_2 = (2 \ln 2 - \frac{1}{3}) - (2 \ln 2 - \frac{1}{2}) + (2 \ln 2 - 1)$$

$$= 2 \ln 2 - (1 - \frac{1}{2} + \frac{1}{3})$$

$$= 2 \ln 2 - \frac{5}{6} \quad (1)$$

$$4 I_3 = 2 \ln 2 - \frac{1}{4} - 3 I_2$$

$$\therefore 4 I_3 = (2 \ln 2 - \frac{1}{4}) - (2 \ln 2 - \frac{1}{3}) + (2 \ln 2 - \frac{1}{2}) + (2 \ln 2 - 1)$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}$$

$$= \frac{7}{12} \quad (1)$$

$$iv) (n+1) I_n = (2 \ln 2 - \frac{1}{n+1}) - (2 \ln 2 - \frac{1}{n}) + \dots + (-1)^{n-1} (2 \ln 2 - \frac{1}{2}) + (-1)^n (2 \ln 2 - \frac{1}{1}),$$

with $n+1$ terms.

Hence if n odd, there is an even number of terms.

$$\boxed{2 \ln 2 - 2 \ln 2 + 2 \ln 2 - 2 \ln 2 + \dots = 0}$$

$$\text{and } (n+1) I_n = -\frac{1}{n+1} + \frac{1}{n} - \dots - \frac{1}{2} + \frac{1}{1}$$

$$= \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1} \quad (1)$$

Also if n is even there is an odd number of terms

$$2 \ln 2 - 2 \ln 2 + 2 \ln 2 - 2 \ln 2 + 2 \ln 2 - \dots = 2 \ln 2$$

$$\text{and } (n+1) I_n = 2 \ln 2 - \frac{1}{n+1} + \frac{1}{n} - \dots + \frac{1}{2} - \frac{1}{1}$$

$$= 2 \ln 2 - (\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n+1}) \quad (1)$$

$$b) i) \tan 4x = \frac{2 \tan 2x}{1 - \tan^2 x}$$

$$= \frac{2 \left(\frac{2t}{1-t^2} \right)}{1 - \left(\frac{2t}{1-t^2} \right)^2} \quad \text{where } t = \tan x \quad (1)$$

$$= \frac{\frac{4t}{1-t^2}}{\frac{(1-t^2)^2 - (2t)^2}{(1-t^2)^2}} \quad (1)$$

$$= \frac{4t(1-t)}{t^4 - 6t^2 + 1}$$

$$8.) b.) ii). \tan x \tan 4x = 1$$

$$\frac{t(4t)(1-t^2)}{t^4 - 6t^2 + 1} = 1 \quad \text{from part (i)}$$

$$4t^2(1-t^2) = t^4 - 6t^2 + 1$$

$$4t^2 - 4t^4 = t^4 - 6t^2 + 1 \quad (1)$$

$$5t^4 - 10t^2 + 1 = 0$$

$$\begin{aligned} iii.) \quad x = 18^\circ \quad \tan x \tan 4x &= \tan 18^\circ \tan 72^\circ \\ &= \tan 18^\circ \cot 18^\circ \\ &= 1 \quad (1) \end{aligned}$$

$$\begin{aligned} x = 54^\circ \quad \tan x \tan 4x \\ &= \tan 54^\circ \tan 216^\circ \\ &= \tan 54^\circ \tan 36^\circ \\ &= \tan 54^\circ \cot 54^\circ \\ &= 1 \quad (1) \end{aligned}$$

$$iv.) \quad 5t^4 - 10t^2 + 1 = 0$$

$$t^2 = \frac{10 \pm \sqrt{100 - 20}}{10}$$

$$= \frac{5 \pm 2\sqrt{5}}{5} \quad (1)$$

Now the roots of the above equation are $\pm \tan 18^\circ$ and $\pm \tan 54^\circ$ and

since $\tan 18^\circ < \tan 54^\circ$ (1)

$$\text{then } \tan 54^\circ = \sqrt{\frac{5 + 2\sqrt{5}}{5}} \quad (1)$$